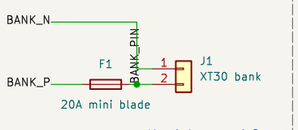
Rev B bank board v2 10S 2.7 V 50 F (27 V 5 F, 1822 J) per-cell TLV431 + PNP parallel clamp

Clamp Vc = Vref(1 + Ra/Rb) + Iref*Ra = 1.24 x (1 + 22.9k/20k) = 2.66 V typ (25 C: about 2.644 – 2.689 V). Charge 26.2 V = 2.62 V/cell average stays below every clamp.

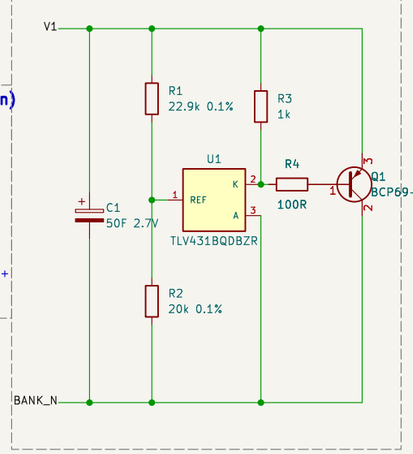
TLV431 SOT-23 (DBZ) pinout from TI SLVS139Z Table 4-1: 1 = REF, 2 = CATHODE, 3 = ANODE. Rbase 100R limits the TLV431 cathode current to about 7 mA (abs max 20 mA).

Only external interface: XT30 (S189). Cells: 50 F 2.7 V, 18 x 40 mm radial, P 7.5 mm – hand soldered (not stocked at JLCPCB). Fuse: 20 A mini blade (1 kA @ 32 V), soldered directly.

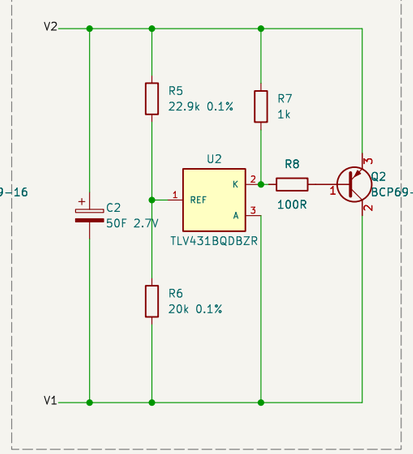
Power port (only external connection)



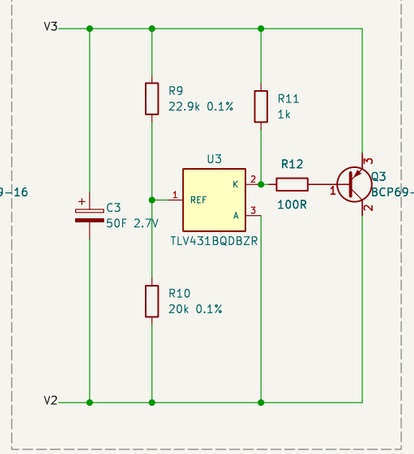
Cell 1



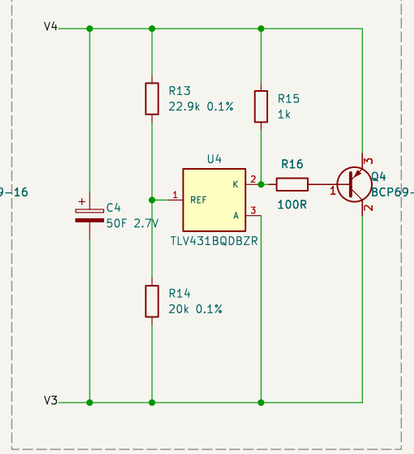
Cell 2



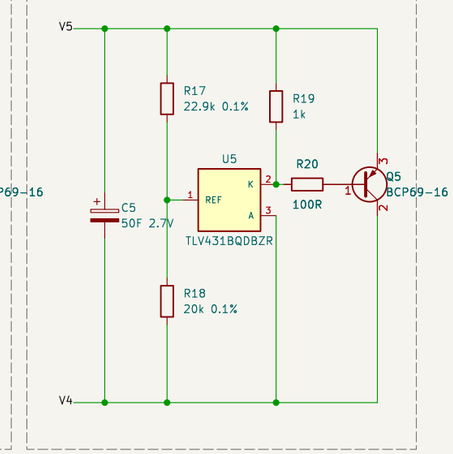
Cell 3



Cell 4



Cell 5



Cell 6

Cell 7

Cell 8

Cell 9

Cell 10

NAGOYA SHACHIHOKO

Sheet: /
File: bank_board.kicad_sch

Title: Rev B bank board v2 – 10S 2.7V 50F, per-cell clamp

Size: A3	Date:	Rev: B-prov
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